BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 102301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. What is the frequency of oscillation if time period is 20 ms ?

(A) 0.02 HZ
(B) 50 HZ
(C) 0.5 HZ
(D) 40 HZ

Q2. The unit of spring constant is :

(A) NC^{-1}
(B) Nm
(C) Nm^{-1}
(D) NS^{-1}

Q3. Which of the following cannot transmit sound ?

(A) Gas
(B) Liquid
(C) Solid
(D) Vacuum

Q4. Index of refraction for diamond is :

(A) 2.417
(B) 1.544
(C) 1.434
(D) 1.333

Q5. Which of the following represents one Ohm ?

(A) JC^{-1}
(B) VA^{-1}
(C) JS^{-1}
(D) WA^{-1}

Q6. The human eye forms the image of an object at its:

(A) Pupil
(B) Iris
(C) Retina
(D) Cornea

Q7. The value of Coulomb's constant (K) depends on:

(A) Value of charges
(B) Material medium
(C) Separation between charges
(D) All of these

Q8. If the distance between two charges is doubled , the force between them becomes :

(A) 1/4th
(B) Four times
(C) Half
(D) Double

Q9. A step-up transformer increases :

(A) Energy
(B) Voltage
(C) Power
(D) Current

Q10. The Boolean expression for NAND gate is :

(A) X= A+ B
(B) X= A+B with bar over (A+B)
(C) X= A.B with bar over (A.B)
(D) X= A with bar over A

Q11. Which of the following information storage device is least use these days ?

(A) Compact disc
(B) Hard drive
(C) Flash drive
(D) Floppy

Q12. What is the number of neutrons in $^{242}_{92}PU$?

(A) 150
(B) 242
(C) 92
(D) 334

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Show that the oscillatory motion of mass-spring system is SHM.

(ii) Why sound produced by a simple pendulum is not heard ?

(iii) State and explain laws of refraction with diagram .

(iv) Briefly describe power of lens and its unit .

(v) How can we protect ourselves during a thunderstorm ?

(vi) What is Kilowatt-hour ? How is it related to joule ?

(vii) Define electric power . Show that P=IV. .

(viii) Why does a compass needle points north ?

(ix) What are logic gates ? Describe the operation of AND gate by drawing its truth table .

(x) Write down four uses of internet in daily life.

(xi) What is meant by beta decay ? Explain briefly .

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] What is intensity level ? Describe the decibels scale for the intensity of different sound levels.

(b) [3] Determine the mass of a vertical mass-spring system if it oscillates with a period of 2s and has the spring constant $K=20Nm^{-1}$.

QIV.

(a) [4] State and explain Coulomb's law .

(b) [3] Three resistors 5 Ω , 10 Ω and 20 Ω are connected in parallel across a 90 V battery. What is the total current flowing in the circuit ?

QV.

(a) [4] What are cathode rays ? How are cathode rays produced ?

(b) [3] A 1.5 m long wire carries a current of 5A at right angle to a uniform magnetic field of 0.04T.What is the force exerted on the wire ?

QVI.

(a) [4] Explain half life of radioactive elements ?

(b) [3] How many neutrons are contained in a gold nucleus $^{197}_{79}Au$